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An Effective Bayesian GLM Approach for IBNR Claim Count Estimation

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Estimating the count of incurred but not reported (IBNR) claims is a fundamental challenge in loss reserving. The Chain Ladder method, a widely used macro-level approach, relies on aggregated claims data and provides a simple framework for reserve estimation. However, it can be inaccurate in many cases as it does not leverage detailed claims information and may introduce biases under certain conditions. To address these limitations, micro-level models have been developed to incorporate individual claim data, capturing claim occurrence and reporting dynamics more effectively. Recent literature has shown that these models outperform the Chain Ladder method in predictive accuracy. Despite their better performance, micro-level models remain largely unused in practice due to their computational complexity and various modeling challenges, such as fitting right-truncated reporting delay distributions and maximizing likelihood functions with interdependent components. The aim of this article is twofold: first, to provide a comprehensive analysis of existing micro-level models, and second, to propose a highly flexible Bayesian framework that builds on the Chain Ladder method while incorporating key micro-level elements such as temporal dynamics and portfolio-level covariates. Crucially, our framework is designed for practical adoption: Thanks to modern probabilistic programming tools like `Stan` and the complete, reproducible code provided, practitioners can readily implement and adapt the model without dealing with complex estimation routines. Through a case study, we demonstrate that our framework not only outperforms the classical Chain Ladder method but also surpasses micro-level models adopted in recent literature, offering a scalable, interpretable, and easily implementable solution for IBNR claim count estimation.

1. INTRODUCTION

Accurate loss reserving plays a pivotal role in the financial health and operational stability of property and casualty (P&C) insurance companies. Reserves must account for all liabilities arising from claims that have occurred, ensuring that sufficient funds are set aside for future payments. Of particular interest are claims that have occurred but are not yet reported—referred to as incurred but not reported (IBNR) claims—since they represent a significant challenge in estimation due to their unobserved nature. In this article, we focus specifically on the estimation of IBNR claim counts, which serve as a crucial first step before reserve estimation.

Traditionally, reserving has been approached using "macro-level" models, such as the Chain Ladder (Mack 1993, 1999) and the Bornhuetter-Ferguson (Bornhuetter and Ferguson 1972) methods. These models rely on aggregated claims data organized into run-off triangles by accident and development years. While these methods are widely used in practice for their simplicity, they come with several disadvantages, including the loss of valuable insights into individual claims data, high parameter uncertainty due to the limited number of observations in aggregated data, and potential bias under specific conditions (Charpentier and Pigeon 2016).

In response, "micro-level" models have emerged, leveraging individual claims data to better capture the intricacies of claim development. The groundwork for micro-level approaches was established by Arjas (1989) and Norberg (1993, 1999), who introduced a stochastic framework based on marked nonhomogeneous Poisson processes. This approach allows for a detailed representation of claim arrival, reporting, and settlement dynamics. Haastrup and Arjas (1996) then implemented the nonhomogeneous Poisson processes model using nonparametric Bayesian statistics. The subject of micro-level reserving remained largely unexplored until Antonio and Plat (2014) presented a case study applying the framework of Arjas (1989) and Norberg (1993, 1999) to an auto-insurance dataset, reigniting interest in micro-level reserving. Recent studies continue to develop models in this tradition, using the probabilistic framework of marked point processes to jointly model claim arrivals and reporting

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delays. The likelihood of the observed data is fitted, and the fitted parameters are then used to estimate IBNR claim counts and reserves.

Despite their promise, micro-level models face several challenges in practice. One of the main challenges is maximizing the likelihood of observed data. This process involves addressing two distinct components: the likelihood for the truncated reporting delay and the likelihood for the claim frequency model of reported claims, which in turn depends on the reporting delay. Thus, maximizing the likelihood poses a formidable task due to the interdependence of these components. Additionally, fitting the right-truncated reporting delay is inherently challenging since empirical data for reporting delay necessitate flexible mixture models (Bücher and Rosenstock 2023), which are not straightforward for right-truncated data. Lastly, these models involve multiple components with modeling assumptions that vary across lines of business due to differing data characteristics, adding to their complexity and limiting their appeal to practitioners who prefer more straightforward methods. Consequently, they are not widely adopted in practice.

The contribution of this article is threefold. First, we provide a critical assessment of the stochastic micro-level reserving framework and its practical applicability. We examine in detail the underlying assumptions and major challenges inherent in such models—particularly issues related to truncated reporting delays and the computational complexity of likelihood-based estimation. This analysis offers a comprehensive evaluation of their limitations, both from a theoretical standpoint and in terms of real-world implementation.

Second, we introduce a simple yet highly efficient extension of the classical Chain Ladder method, formulated within a Bayesian framework, that incorporates a temporal structure and covariates reflecting the risk characteristics of the insured portfolio. Although aggregate in nature, the proposed model is sufficiently flexible to capture key features often addressed only by more complex micro-level approaches. One of our central aims is to demonstrate that this natural and interpretable extension of Chain Ladder can consistently outperform a broad range of micro-level models from the literature, which have been regarded as highly accurate for reserve prediction.

By focusing on aggregate claims count data, our framework maintains computational efficiency and interpretability. The Bayesian formulation enables direct inference on full predictive distributions without resorting to bootstrapping, while naturally accommodating potential overparameterization. Most importantly, the model is designed for ease of implementation and adaptation in practice—avoiding the prohibitive complexity and shifting model assumptions that often render fully granular micro-level approaches inaccessible to practitioners, particularly across diverse lines of business.

Bayesian methods have been widely applied in the context of loss reserving, particularly for implementing the overdispersed Poisson (ODP) model and its extensions. For comprehensive overviews of Bayesian reserving approaches, see Meyers (2015) and Gao et al. (2018). In most of these models, time-dependent parameters are treated as fixed effects. In this article, we extend this framework by introducing temporal structures on the parameters and by incorporating covariates that capture the risk characteristics of the entire portfolio, allowing these covariates to influence both occurrence and reporting dynamics. This enhances the model's ability to capture systematic trends and variations in claim development over time. In addition, we demonstrate how practitioners can readily implement a wide range of temporal specifications in `Stan`, requiring only minimal code adjustments. This makes our framework not only flexible and powerful but also highly accessible for industry use, bridging the gap between academic advances and practical reserving applications.

Third, through an extensive case study using an auto insurance dataset from a European country, we conduct a carefully designed comparison between a broad class of micro-level models, the classical Chain Ladder method, and our proposed extension. The results clearly illustrate the advantages of the proposed model, both in terms of predictive accuracy and practical applicability.

The article is structured as follows. In Section 2, we review the micro-level model proposed by Norberg (1993) and briefly discuss both the theoretical and practical limitations of these models and their application to real-world data. In Section 3, we present our proposed Bayesian extension to the chain ladder model. In Section 4, we apply the proposed model to an auto insurance dataset from a European country. Section 5 offers a brief discussion of the trade-offs between different reserving approaches for IBNR claim counts. Finally, we conclude in Section 6.

2. STOCHASTIC MICRO-LEVEL MODELS: OVERVIEW AND CHALLENGES

This section starts by reviewing the continuous-time stochastic micro-level framework, originally proposed by Arjas (1989) and Norberg (1993, 1999), which has served as the foundation for various micro-level reserving models. We review the key contributions that have shaped the evolution of these methods, with a particular emphasis on critically examining their core modeling assumptions and the ways in which they have been adapted for estimating IBNR claims counts and reserves.

Despite their theoretical appeal, such models have seen little adoption in practice, as evidenced by the survey in Francis (2016), mainly due to their complexity compared with traditional reserving methods. Even if computational and implementation barriers were removed, the way these models are typically specified and fitted introduces statistical limitations that may affect predictive accuracy and reliability. The discussion that follows presents both the framework itself and the main challenges encountered when applying it to claims reserving.

2.1. An Overview

The development of a claim is described through three random variables $(T, U, \boldsymbol{Z})$, where T denotes the occurrence time of the claim, U represents its reporting delay, and $\boldsymbol{Z}$ characterizes the development process following claim reporting, including information such as the type of the claim, the number of payments for the claim, the sizes of these payments, and so on. Chronologically ordered by their arrival, $\{(T_j, U_j, \boldsymbol{Z}_j), j = 1, 2, ...\}$ constitutes the claim history process for the portfolio. Additionally, the total claim count process is defined as

$$N(t) = \sum_{j=1}^{\infty} \mathbb{1}\{T_j < t\},$$

where $\mathbb{1}\{.\}$ is the indicator function. Arjas (1989) and Norberg (1993, 1999) proposed modeling $N(t)$ using a marked nonhomogeneous Poisson process with intensity $\lambda(t)$. The marks $\{(U_1, \boldsymbol{Z}_1), (U_2, \boldsymbol{Z}_2), ...\}$ are assumed to be independent and share a common density function given by

$$f_{U,\boldsymbol{Z}|t}(u, z) = f_{U|t}(u) f_{\boldsymbol{Z}|u,t}(z),$$

where $f_{U|t}$ denotes the conditional density of the reporting delay U given $T = t$, and $f_{\boldsymbol{Z}|u,t}$ denotes the conditional density of $\boldsymbol{Z}$ given $(U = u, T = t)$. The corresponding processes of reported and IBNR claims with respect to a reserve valuation date τ are given by

$$N^{\mathrm{r}}(t; \tau) = \sum_{j=1}^{\infty} \mathbb{1}\{T_j < t,\ \ T_j + U_j \leq \tau\}$$

and

$$N^{\mathrm{ibnr}}(t; \tau) = \sum_{j=1}^{\infty} \mathbb{1}\{T_j < t,\ \ T_j + U_j > \tau\},$$

for $0 \leq t \leq \tau$. The first expression counts the number of claims that occurred before time t and were reported by the valuation date τ, while second counts the number of claims that occurred before time t but are still unreported at τ.

By the thinning property, the reported claim process $\{N^{\mathrm{r}}(t; \tau) : 0 \leq t \leq \tau\}$ is a marked nonhomogeneous Poisson process with intensity $\lambda(t) F_{U|t}(\tau - t)$, and marks distributed according to the conditional density

$$f^{\mathrm{r}}_{U,\boldsymbol{Z}|t}(u, z) = \frac{f_{U|t}(u)\, \mathbb{1}\{u \leq \tau - t\}}{F_{U|t}(\tau - t)} f_{\boldsymbol{Z}|u,t}(z),$$

where $F_{U|t}$ denotes the conditional distribution of the reporting delay U given $T = t$. Similarly, the unreported (IBNR) claim process $\{N^{\mathrm{ibnr}}(t; \tau) : 0 \leq t \leq \tau\}$ is a marked nonhomogeneous Poisson process with intensity $\lambda(t)(1 - F_{U|t}(\tau - t))$, and marks with density

$$f^{\mathrm{ibnr}}_{U,\boldsymbol{Z}|t}(u, z) = \frac{f_{U|t}(u)\, \mathbb{1}\{u > \tau - t\}}{1 - F_{U|t}(\tau - t)} f_{\boldsymbol{Z}|u,t}(z).$$

Since only reported claims are observed by the valuation date τ, the likelihood is based on the observed claims

$$\mathcal{O} = \{t^{\mathrm{r}}_j, u^{\mathrm{r}}_j, z^{\mathrm{r}}_j; j = 1, ..., N^{\mathrm{r}}(\tau)\},$$

and decomposes into three components:

$$\mathcal{L}(\mathcal{O}) = \underbrace{\left\{\prod_{j\geq 1} \lambda(t_j^{\mathrm{r}}) F_{U|t}(\tau - t_j^{\mathrm{r}})\right\} \exp\left(-\int_0^{\tau} w(t)\lambda(t) F_{U|t}(\tau - t)dt\right)}_{\text{Claim Frequency}} \times \underbrace{\prod_{j\geq 1} \frac{f_{U|t}(u_j^{\mathrm{r}})}{F_{U|t}(\tau - u_j^{\mathrm{r}})}}_{\text{Reporting Delay}} \times \underbrace{\prod_{j\geq 1} f_{\mathbf{Z}|u,t}(\mathbf{z}_j^{\mathrm{r}})}_{\text{Claim Development}}, \tag{1}$$

where $w(t)$ denotes exposure at time t. Several key aspects of the likelihood function (1) are worth highlighting. First, the *Claim Frequency* term captures the assumed process governing the arrival of claims. Second, the *Reporting Delay* component accounts for the right-truncated nature of the observed data, as only claims reported within the observation window are included. Importantly, the claim frequency and reporting delay components are interdependent—the likelihood function makes it explicit that the frequency of observed claims depends on the distribution of reporting delays. Finally, the *Claim Development* component is right-censored, as some claims remain unsettled by the end of the observation period, leaving their development histories only partially observed.

This general framework has been adopted in various studies, with adaptations falling into three main directions. First, some follow the original structure while specifying a particular form for the time-dependent intensity $\lambda(t)$, such as a piecewise constant function (Antonio and Plat 2014) or a constant function, reducing the model to a homogeneous Poisson process (Pigeon et al. 2013, 2014; Godecharle and Antonio 2015). Second, others extend the nonhomogeneous Poisson formulation to a Cox process, allowing $\lambda(t)$ to vary stochastically over time to capture temporal dependence and calendar-year effects that are ignored by the nonhomogeneous Poisson process, as in the hidden Markov model (HMM)-based approach of Badescu et al. (2016, 2019). Third, more recent work applies the framework at the policyholder level, letting model components depend on individual risk characteristics; for instance, Fung et al. (2021) assumed a homogeneous Poisson process for each policyholder with intensities driven by individual covariates, while Abdelrahman et al. (2025) extended the HMM-based Cox model of Badescu et al. (2016, 2019) to incorporate such covariates, showing substantial gains in IBNR claim count estimation.

Once the model components for claim frequency, reporting delay, and claim development are fitted, the number of IBNR claims can be estimated using the implied intensity of the unreported claims process. This is typically done by simulating from the corresponding Poisson distribution. In Cox process extensions (e.g., Badescu et al. 2016; Abdelrahman et al. 2025), the thinning property ensures the same approach applies, with simulation based on the stochastic intensity. Reserve estimation then proceeds by simulating the full development of each IBNR claim—for example, number and size of payments—using the fitted postreporting distributions.

2.2. Challenges and Limitations

2.2.1. Likelihood Fitting

The likelihood (1) has the same structure across the continuous-time implementations already described. In all contributions following the continuous-time framework—except Wahl et al. (2019)—the interdependent claim frequency and reporting delay components are estimated via a two-step maximization: The reporting delay component is first estimated independently, and its fitted parameters are then used to estimate the claim frequency component. While computationally convenient, this approach ignores the shared structure between components and can lead to suboptimal parameter estimates. In models with time-dependent claim arrival intensity, such as nonhomogeneous Poisson or Cox processes, the resulting distortions can be substantial, particularly for IBNR claim count predictions.

To illustrate, consider the nonhomogeneous Poisson process with piecewise constant intensity as in Antonio and Plat (2014). The claim arrival intensity is $\lambda(t) = \lambda_l, \quad \text{for } d_{l-1} \leq t < d_l, \quad l = 1, ..., T,$ where $d_0 = 0$ and $d_T = \tau$. Let $w(t) := w_l$ be the exposure in interval $[d_{l-1}, d_l)$, and N_l^{r} the number of reported claims therein. The MLE of λ_l is

$$\hat{\lambda}_l = \frac{N_l^{\mathrm{r}}}{w_l \int_{d_{l-1}}^{d_l} F_{U|t}(\tau - t)dt}.$$

In the final interval $[d_{T-1}, d_T)$, claims have the shortest reporting window, so the IBNR proportion is typically largest. The integral $\int_{d_{T-1}}^{d_T} F_{U|t}(\tau - t)dt$ represents the probability of a claim from this interval being reported before τ. Underestimating this probability inflates $\hat{\lambda}_T$, as the denominator in the MLE becomes too small.

For IBNR estimation, unreported claims in the final interval follow

$$\text{Poisson}\left(\lambda_T\left[1 - \int_{d_{T-1}}^{d_T} F_{U|t}(\tau - t)dt\right]\right).$$

If both λ_T and the unreported fraction $1 - \int_{d_{T-1}}^{d_T} F_{U|t}(\tau - t)dt$ are overestimated, the bias compounds, yielding substantial overprediction. Conversely, overestimating the reporting probability produces downward bias. This sensitivity is exacerbated when reporting patterns exhibit high random variability.

2.2.2. The Right-Truncated Reporting Delay

A further complication is that the reporting delay distribution is inherently right-truncated. Some studies ignore truncation, which simplifies estimation but biases the tail behavior and underestimates long delays. Another strategy is *back-censoring* (e.g., Fung et al. 2021), in which an earlier cutoff date is chosen so that all claims within the observation window have complete reporting delay information. While computationally straightforward, this approach is not optimal as it ignores reporting delays of recent accident periods. Another approach is to use time-reversal techniques, which convert right-truncated data into left-truncated form, allowing the use of standard survival methods (e.g., Hiabu 2017). Finally, more recent approaches adjust directly for truncation, such as the logit-weighted reduced mixture of experts model (Fung et al. 2022) or hybrid parametric–neural network models with heavy-tail components (Bücher and Rosenstock 2023). While these methods improve robustness, they may pose challenges in terms of practical implementation for actuarial practitioners.

2.2.3. Computational Challenges

Even after fitting the reporting delay and simplifying the estimation via a two-step maximization approach, the *Claim Frequency* component still requires evaluating integrals of the form

$$\int_0^\tau w(t)\lambda(t)F_{U|t}(\tau - t)dt.$$

This step can become a major computational bottleneck, particularly when $\lambda(t)$ or $F_{U|t}$ is modeled with flexible structures. The challenge is further amplified in an individual-level setting with policyholder covariates, because analogous integrals must be evaluated repeatedly for each policyholder.

2.2.4. Discrete-Time Alternatives

The difficulties just described have motivated discrete-time micro-level alternatives. These retain the individual-policyholder modeling unit but aggregate events onto a fixed time grid (e.g., weeks or months), preserving a hierarchical structure in which claim occurrence is modeled first and reported claims are allocated across delay bins. The allocation is naturally represented by a multinomial structure, which implicitly captures the delay distribution and removes the need to specify it separately; consequently, the full likelihood can often be maximized in a single step, avoiding the two-stage procedures common in continuous-time formulations.

For example, Crevecoeur et al. (2023) proposed an expectation-maximization (EM) algorithm for Poisson occurrence processes, while Abdelrahman et al. (2025) extended this to Poisson–HMM processes. Notably, Abdelrahman et al. (2025) formulated their HMM-based model in both continuous- and discrete-time settings at the individual level. Their case study found that, despite some information loss from discretization, the discrete-time model outperformed the continuous-time alternative due to the joint modeling of claim occurrence and reporting.

Importantly, while discrete-time aggregation may resemble macro-level methods when applied at the portfolio level, the modeling unit is different: Here, *discrete-time micro-level models* refers to implementations at the individual policyholder level. Although discrete-time micro-level models address many issues with their continuous-time counterparts, they often rely on EM-based estimation and can still be computationally demanding, motivating the search for simpler yet flexible methods that remain practical for industry use.

3. A GLM-BASED FRAMEWORK FOR IBNR CLAIMS COUNT

Despite the complexities associated with micro-level models, as discussed in the previous section, empirical evidence shows that they generally outperform traditional reserving approaches such as the Chain Ladder method in estimating IBNR claims count. In this section, we propose a generalized linear model (GLM) framework that operates at the aggregate level while incorporating micro-level flavor, aiming to outperform both micro-level models and traditional approaches in estimating IBNR claims count, yet without the complexities inherent to full micro-level implementations.

While recent literature increasingly explores machine learning and neural networks applications in reserving, most applications focus on reported but not settled (RBNS) claims rather than IBNR. As outlined in Bücher and Rosenstock (2023), the few studies that do focus on IBNR claims count estimation typically rely on either a frequency–severity approach or a Chain Ladder-based method, often incorporating machine learning or neural network models applied to aggregated claims count data (e.g., Lindholm et al. 2020; Delong et al. 2022). However, applying such advanced models to aggregated data presents several limitations, including data scarcity, lack of interpretability, and potential overfitting. Moreover, as concluded in Jamal et al. (2018), these methods do not necessarily provide better estimates and face challenges when data is limited or imbalanced. Rather than relying on such complex and potentially opaque models, we propose a Bayesian GLM framework that offers a more transparent, flexible, and statistically sound approach to IBNR claims count estimation.

3.1. The Model

We develop our model as an extension of the classical Chain Ladder framework, preserving its aggregate data structure while enhancing its flexibility to incorporate additional sources of variation. As highlighted in Renshaw and Verrall (1998), the Chain Ladder method for claims count data can be interpreted as a GLM, where the number of claims $N_{t,d}$ occurring in accident period t and reported with delay d is modeled as

$$N_{t,d} \sim \text{Poisson}(\lambda_{t,d}), \quad \text{with} \quad \lambda_{t,d} = \alpha_t \beta_d, \tag{2}$$

subject to the constraint $\sum_{d=0}^{D} \beta_d = 1$, with $t = 1, ..., T$ and $d = 0, ..., D$, where $N_{t,d}$ is only observed if $t + d \leq T$.

Traditionally, the Chain Ladder is applied to annually aggregated data with $T = D$, and α_t and β_d are treated as fixed effects. However, the formulation is more flexible: The level of aggregation can be chosen freely (e.g., monthly or quarterly). Empirical evidence from Tee and Käärik (2022) shows that the Chain Ladder performs better at finer levels of aggregation, as more information about the development process is retained. At such levels, it becomes natural to impose temporal structure on the time effects, which helps stabilize estimation and capture the smooth evolution of claims over time.

To this end, the classical model can be extended in two key directions:

- **Temporal structure.** Rather than modeling α_t and β_d as fixed effects—an approach suitable primarily when data are aggregated annually—we treat them as temporally structured random effects. This enables the model to capture smooth trends and serial dependence in claim development across accident and delay periods, while borrowing strength across neighboring time points.
- **Incorporation of portfolio characteristics.** To integrate portfolio-level risk characteristic while maintaining a macro-level framework, we include summary statistics of policyholder-level covariates as predictors in the model. Importantly, we allow these covariates to influence the reporting delay structure. This reflects the fact that policyholder characteristics may influence not only the frequency of claims but also the way claims are distributed across different reporting delays.

These extensions are motivated by the same reasons that micro-level models often outperform the traditional Chain Ladder: They make better use of granular information and incorporate heterogeneity in claim development. By embedding these features into an aggregate framework, we aim to capture much of the predictive gain of micro-level approaches while preserving the practicality of macro-level modeling.

Formally, we assume

$$N_{t,d} \sim \text{NegBin}(w_t \lambda_{t,d}, \phi), \tag{3}$$

with

$$\mathbb{E}[N_{t,d}] = w_t \lambda_{t,d}, \quad \text{Var}(N_{t,d}) = w_t \lambda_{t,d} \left(1 + \frac{w_t \lambda_{t,d}}{\phi}\right), \tag{4}$$

where we allow for overdispersion by adopting a negative binomial likelihood. The term w_t represents an exposure offset, ensuring that the model accounts for variations in the number of insureds in accident period t.

The log-mean claim count per unit exposure is modeled as

$$\log(\lambda_{t,d}) = \mu + \alpha_t + \beta_d + \gamma_{t,d} + \langle \boldsymbol{\delta}_d, Z_t \rangle + \langle \boldsymbol{\eta}_d, X_t \rangle, \tag{5}$$

where μ denotes the baseline log-mean per unit of exposure; α_t is a temporally structured random effect capturing the evolution of claim frequency over accident periods; β_d is a delay-specific effect describing the reporting delay mechanism, which can also be given temporal structure if D is sufficiently large; $\gamma_{t,d}$ captures accident–delay interactions through a temporally structured random effect; Z_t are time-specific covariates such as calendar month, holidays, or macroeconomic indicators, with delay-varying coefficients $\boldsymbol{\delta}_d$; and X_t are summary statistics of policyholder-level covariates (e.g., average policyholder age), with delay-varying coefficients $\boldsymbol{\eta}_d$.

This formulation is deliberately general: The relevance of each component depends on the data and application, and terms can be omitted if unnecessary. For example, including X_t is important when portfolio risk characteristics change over time, while incorporating Z_t such as month of occurrence is crucial for capturing seasonal patterns in occurrence or reporting. In traditional practice, such seasonality is often addressed by aggregating data annually before applying methods like Chain Ladder, which sacrifices granularity. Our approach allows finer time units (e.g., monthly) while explicitly modeling seasonality, thus retaining more information.

On the other hand, modeling α_t, β_d, and $\gamma_{t,d}$ as time-series processes rather than as independent fixed effects has several advantages. First, it allows information to be shared across neighboring periods, stabilizing estimates in sparse cells and reducing the effective number of free parameters. This reduction mitigates overfitting while still capturing smooth trends, seasonality, and occasional level shifts. Second, the dynamic specification defines the latent paths for all t (and $t+d$), so the model can forecast into the lower triangle in a principled way. Importantly, simply placing i.i.d. priors such as $\gamma_{t,d} \sim \mathcal{N}(0, \tau^2)$ for all t, d without a temporal link does not borrow information across time; posterior draws for unobserved cells then default toward the prior mean (with variance driven by τ^2) and are essentially uninformed by the data. A temporal structure is what transmits information from observed to unobserved cells and produces data-informed forecasts with coherent uncertainty.

While the model structure we propose is not inherently Bayesian—frequentist estimation would also be possible—we adopt a fully Bayesian estimation framework. This choice is motivated by several practical advantages of Bayesian inference in this context, drawing inspiration from recent developments in nowcasting in disease surveillance analysis (e.g., Bastos et al. 2019; McGough et al. 2020):

1. **Regularization through priors:** Bayesian modeling accounts for parameter uncertainty through prior distributions, effectively regularizing the estimation process and mitigating overfitting—a common concern in GLM-based models.
2. **Flexible and smooth temporal modeling:** The Bayesian formulation accommodates a wide range of temporal structures (random walks of different orders, AR processes, splines, Gaussian processes) simultaneously on accident, delay, and interaction effects. Using probabilistic programming (e.g., `Stan`, `NIMBLE`, `JAGS`) avoids writing custom Kalman filters or optimizers for each specification; one only modifies the prior/latent process definitions and the sampler handles inference.
3. **Direct predictive sampling:** IBNR prediction is obtained by posterior predictive sampling from the likelihood using draws of $\lambda_{t,d}$, ϕ, and the latent temporal effects extended into $t+d > T$. This yields coherent predictive distributions without additional ad hoc steps such as bootstrapping the residual triangle.

This extension of the GLM-based Chain Ladder model is intuitive yet powerful. As shown in our case study, it can match or exceed the predictive performance of more complex micro-level approaches while being significantly more practical and easier to implement in real-world reserving applications.

3.2. Estimation and Prediction in `Stan`

We estimate the model using the programming language `Stan`, interfaced through the `rstan` package in R. `Stan` performs full Bayesian inference via Hamiltonian Monte Carlo (HMC), an efficient Markov-chain Monte Carlo (MCMC) method that allows us to draw samples from the joint posterior distribution of all model parameters (Carpenter et al. 2017).

The posterior distribution includes samples of the baseline mean μ, the temporally structured effects α_t, β_d, and $\gamma_{t,d}$ for all $t = 1, ..., T$ and $d = 0, ..., D$, the delay-varying regression coefficients $\boldsymbol{\delta}_d$ and $\boldsymbol{\eta}_d$, and the overdispersion parameter ϕ. Given

these samples, we compute the corresponding posterior draws of the mean claim counts $\lambda_{t,d}$ for all combinations such that $t + d > T$ and $t + d \leq T + D$.

The IBNR claim count is then defined as the sum of the unobserved counts $N_{t,d}$ over all (t, d) such that $t + d > T$. For each posterior draw, we simulate these $N_{t,d}$ values from the negative binomial likelihood using the drawn values of $\lambda_{t,d}$ and ϕ. This yields a full posterior sample of the IBNR claim count, which naturally reflects both parameter uncertainty and process variability. A point estimate of the IBNR claim count can be obtained as the posterior mean or median, and credible intervals follow directly from the posterior sample.

A detailed guide for fitting the model in `Stan` and obtaining the posterior distribution of the IBNR claims count is provided in the Appendix.

3.2.1. Remarks

1. GLM-based reserving models have long been studied in the context of the total reserve, typically focusing on claim severities and treating accident and delay effects as fixed parameters, sometimes with the inclusion of calendar year or macroeconomic covariates (see Taylor and McGuire (2016) for a review). In this literature, no distinction is made between RBNS and IBNR components. State-space models with fixed dynamic structures, such as local trends or AR(1) processes applied to development factors, have also been proposed (see Chukhrova and Johannssen (2017) for a review). These approaches, however, restrict the form of temporal dependence and require bespoke estimation routines tailored to each specification. By contrast, our formulation—focused on IBNR claim count estimation—is deliberately more general: Accident, delay, and interaction effects can each be endowed with a wide variety of temporal priors, including random walks of different orders, autoregressive processes, splines, or Gaussian processes, all within a single GLM framework. A further advantage is practical: By adopting a Bayesian setting and implementing the model in a probabilistic programming tool such as `Stan`, inference becomes automatic regardless of the chosen temporal prior. This frees practitioners from the technical challenges of estimation and makes flexible Bayesian temporal modeling directly accessible in routine actuarial work.
2. Although calendar year effects are more commonly associated with claim severity (e.g., due to inflation), they may also play a role in claim counts if reporting practices or regulatory environments evolve over time. Our framework can readily incorporate such effects by adding a term κ_{t+d}, with the same flexibility to impose temporal structure as for accident and delay effects.
3. The use of `Stan` remains limited in the reserving literature and in actuarial science more broadly. Existing applications include Gao et al. (2018), who demonstrate how standard reserving models can be implemented in `Stan`, and Gao and Meng (2018), who develop a Bayesian spline model with random loss ratio effects. Our work contributes to this line of research by applying probabilistic programming to claims count reserving (i.e., IBNR claims), and by showing how actuaries can exploit `Stan` to implement flexible temporal structures within a coherent GLM framework.

4. CASE STUDY: REAL-LIFE DATA

To assess the performance of our framework, we analyze a real-life auto insurance dataset from a European insurer that exhibits noticeable variability in claim frequency and reporting delays. We compare our Bayesian approach against the classical Chain Ladder method and a range of micro-level models from the literature. In light of the discussion in the previous sections, we also highlight the limitations of micro-level approaches in this context.

4.1. Data Description

In this study, we analyze an auto insurance dataset from a European insurer, previously examined in Abdelrahman et al. (2025). The dataset comprises physical damage policies issued between January 2009 and December 2017, covering a total of 293,709 one-year policies and 106,187 reported claims. Each policy record includes information about the policyholder and vehicle. The dataset also contains claim details, including the date of occurrence, reporting date, and payment history.

4.1.1. Reporting Delay

Between January 2009 and December 2017, nearly all claims were reported within two years of occurrence, with only two exceptions. This suggests that all claims occurring before 2016 are highly likely to be reported. The majority of claims (approximately 82%) were reported in the same month they occurred, with over 99% being reported within five months.

Delays beyond 10 months were rare, affecting only a small fraction of claims. Table 1, reproduced from Abdelrahman et al. (2025), summarizes the distribution of reporting delays.

4.1.2. Claim Frequency

The dataset reveals substantial fluctuations in claim frequency over time. While exposure (i.e., the number of active policies) followed a distinct trend—rising steadily until late 2012, declining until early 2015, and then increasing again—the claim frequency per exposure exhibited more irregular variation. Notably, the claim frequency ranged between 0.023 and 0.045 claims per exposure. Figure 1, reproduced from Abdelrahman et al. (2025), illustrates these trends in exposure and claim frequency per exposure over time.

4.2. IBNR Claims Count Estimation Models

To evaluate the effectiveness of our proposed framework, we compare it against several benchmark methods, including classical reserving techniques, variations of our proposed model, and micro-level models from the literature. The models considered are as follows:

1. **Chain Ladder method:** We aggregate claims count data into a monthly run-off triangle and apply the classical Chain Ladder method to estimate the number of IBNR claims. This choice of monthly granularity ensures consistency with the other discrete-time models considered in our analysis.
2. **Micro-level models from literature:** We consider both continuous-time and discrete-time micro-level reserving models, where the claim arrival process is modeled at the policyholder level, incorporating individual risk characteristics as covariates.
   - Continuous-time models:
     - **HMM-based Cox process:** This corresponds to the "Continuous Model" in Abdelrahman et al. (2025), where it was previously fitted to the same dataset. It is an HMM-based Cox process that assumes a shared hidden Markov process governing claim arrival processes across all policyholders within the portfolio. The HMM serves as an

TABLE 1
Reporting Delay for Claims Occurring Before 2016 (Reproduced from Abdelrahman et al. (2025))

Delay (months)	0	1	2	3	4	5	6	7	8	9	10+
Number of claims	66475	11617	1580	642	342	192	141	105	80	52	130
Percent of total (%)	81.71	14.28	1.94	0.79	0.42	0.24	0.17	0.13	0.10	0.06	0.16
Cumulative percent (%)	81.71	95.99	97.93	98.72	99.14	99.38	99.55	99.68	99.78	99.84	100.0

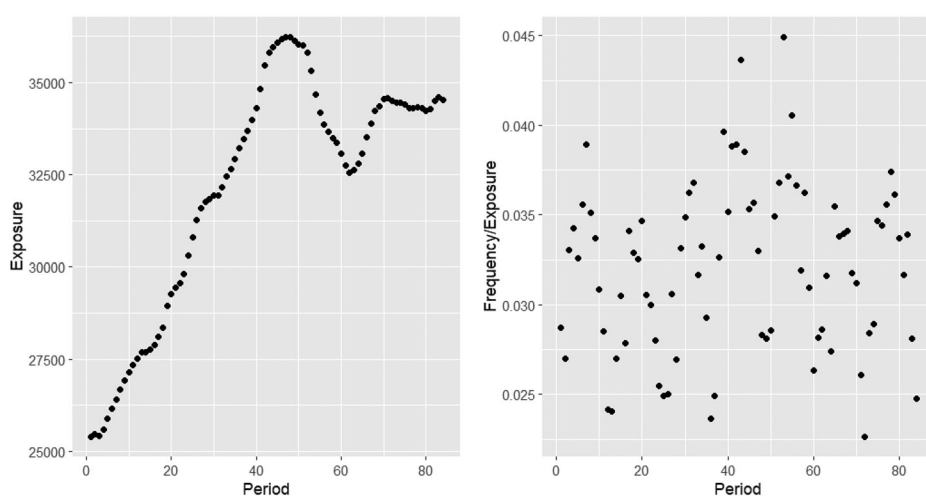


FIGURE 1. Exposure & Claim Count Per Exposure Between January 2009 and December 2015 (Reproduced from Abdelrahman et al. (2025)).

external environmental dynamic that influences all policyholders simultaneously. Given a state of the HMM, the claim arrival intensity for each policyholder is modeled as a constant, determined by their risk attributes through a regression function. Further details on the model fitting procedure can be found in Abdelrahman et al. (2025).

– **Poisson process:** A standard continuous-time Poisson process incorporating covariates. This serves as a special case of the Cox continuous model with only one hidden state, meaning claim arrival intensities depend solely on policyholder characteristics without external latent influences. This baseline model was employed in Fung et al. (2021).

As previously discussed, the likelihood of observed data for both continuous-time models consists of two components: the *Claim Frequency* and the *Reporting Delay*. Before fitting the claim frequency component, we first estimate the distribution of the reporting delay, which is then utilized to adjust the claim counts accordingly. A key challenge in this process is the right truncation of the reporting delay, which can bias the fitted distribution.

As done in Abdelrahman et al. (2025), we provide the continuous-time models an advantage by fitting the reporting delay using all claims that occurred before the reserve valuation date, including those that have not yet been reported. This mitigates the right truncation issue, allowing for a more accurate fit. While this approach is infeasible in practice, it ensures that any performance gap between these models and our proposed methods is not due to an inadequate treatment of right truncation. The reporting delay distribution is modeled using the log-logistic proportional hazard regression, as employed in Abdelrahman et al. (2025), where it demonstrated a good fit based on the Cox-Snell residual analysis.

- **Discrete-time models:** For the discrete-time micro-level models, we recall from Section 2.2 that the claim occurrence and reporting process can be discretized into period-based counts, where the reporting delay structure is modeled through a multinomial distribution. In this setting, we consider the following discretized versions of the continuous-time models.

  – **HMM-based Cox process:** This is the discrete-time counterpart of the Cox continuous model, corresponding to the "multinomial model" in Abdelrahman et al. (2025).
  – **Poisson process:** This corresponds to the discrete-time Poisson model in Crevecoeur et al. (2023).

The estimation procedures for these models follow the methodologies detailed in Abdelrahman et al. (2025) and Crevecoeur et al. (2023), respectively. In these discrete-time models, the claim occurrence and reporting structure are defined using monthly periods. Additionally, we set a maximum reporting delay of $D = 9$ months, meaning that claims occurring in period t are assumed to be reported by $t + 9$. This choice is motivated by empirical evidence showing that 99.9% of claims are reported within this timeframe (as indicated in Table 1). Furthermore, for $d > 9$, the GLM regression fails to converge due to the sparsity of claims reported beyond this period.

Similarly, for the HMM-based models, the hidden Markov processes are defined over monthly intervals, and the number of states is set to four, as this choice yields the lowest Bayesian information criterion (BIC) among alternative specifications. Note that we include the HMM-based models from our earlier work, as they demonstrated strong predictive performance for IBNR claims count estimation on this dataset when compared to a wide range of alternative approaches. Moreover, they operate at the policyholder level and incorporate a flexible temporal structure, allowing them to capture time-dependent dynamics in the claim arrival process.

3. **Inverse probability weighting (IPW) method:** Several studies (e.g., Fung et al. (2022) and Calcetero-Vanegas et al. (2023)) have shown that an estimate of the IBNR claims count can be obtained solely from the reporting delay distribution, and is given by

$$\sum_{i=1}^{N^{\mathrm{r}}(\tau)} \frac{1 - F_{U|t}(\tau - t_i; \boldsymbol{x}_i)}{F_{U|t}(\tau - t_i; \boldsymbol{x}_i)}$$

where $N^{\mathrm{r}}(\tau)$ represents the number of reported claims by the valuation date τ, and $\boldsymbol{x}_i$ denotes the characteristics of the i-th reported claim. This fraction is equivalent to the inverse probability weighting formulation in sampling theory, as indicated by Calcetero-Vanegas et al. (2023). We refer to this approach as the IPW method. Similar to the continuous-time micro-level models, we fit the reporting delay regression using a log-logistic proportional hazard model, and using all claims that occurred before the reserve valuation date, including those that have not yet been reported. This mitigates the right truncation issue, ensuring a more accurate estimation of reporting probabilities.

4. **Proposed model:** We fit the proposed model introduced in Section 3 in two variants:
   - **Bayesian baseline model:** A version of our framework where the regression coefficients $\boldsymbol{\delta}$ and $\boldsymbol{\eta}$ do not vary with the reporting delay d. Here, the delay structure is determined solely by the effect β_d, without additional influence from covariates.
   - **Bayesian full model:** The complete version of our model, where the regression coefficients $\boldsymbol{\delta}$ and $\boldsymbol{\eta}$ vary with the reporting delay d. This allows covariates summarizing the portfolio's risk characteristics as well as time-dependent covariates to influence the reporting delay structure dynamically, capturing more complex interactions.

As with the discrete-time micro-level models, we set a maximum reporting delay of $D = 9$ months and use monthly aggregation. The model components follow the structure introduced earlier (see Equation 5), with the following temporal structures and prior specifications:

- **Dispersion parameter ϕ:** We assign an Exponential(1) prior to ϕ, a conventional weakly informative choice that favors smaller values while leaving sufficient mass on larger values to allow for substantial overdispersion relative to the Poisson model.
- **Intercept μ:** The baseline log-mean count per unit exposure μ is given a Normal$(0, 2)$ prior. This weakly informative prior is centered at zero and wide enough to comfortably cover the plausible range of log claim rates in the data.
- **Regression coefficients $\boldsymbol{\delta}_d$, $\boldsymbol{\eta}_d$:** Independent Normal$(0, 1)$ priors are assigned to regression coefficients. This is a standard choice in regression modeling that regularizes estimation without imposing strong prior beliefs.
- **Delay effects β_d:** Delay-specific fixed effects are modeled with independent Normal$(0, 1)$ priors. Since the number of delay periods is relatively small ($D = 9$), no additional temporal structure is imposed.
- **Temporal effects α_t:** Accident-period effects α_t follow a first-order random walk,

$$\alpha_1 \sim \text{Normal}(0, \sigma_\alpha), \quad \alpha_t \sim \text{Normal}\ (\alpha_{t-1}, \sigma_\alpha) \quad \text{for } t \geq 2,$$

  with $\sigma_\alpha \sim \text{HalfNormal}(0, 1)$. This prior allows for smooth temporal evolution while remaining weakly informative about the degree of variation.

- **Time-delay interactions $\gamma_{t,d}$:** Each delay-specific path $\{\gamma_{t,d}\}_t$ is also modeled as a first-order random walk,

$$\gamma_{1,d} \sim \text{Normal}(0, \sigma_\gamma), \quad \gamma_{t,d} \sim \ \text{Normal}(\gamma_{t-1,d}, \sigma_\gamma) \quad \text{for } t \geq 2,$$

  with $\sigma_\gamma \sim \text{HalfNormal}(0, 1)$. This allows for delay-specific deviations over calendar time, while encouraging moderate smoothing.

All additive components α_t, β_d, and $\gamma_{t,d}$ are constrained to sum to zero over their respective indices to ensure identifiability of the intercept μ. The chosen priors are conventional weakly informative specifications, ensuring stable estimation while leaving the results primarily driven by the data. Example `Stan` code implementing these priors is provided in Appendix A.

While more complex temporal structures could be imposed on α_t and $\gamma_{t,d}$, we begin with the most basic specification—a first-order random walk. This serves as a baseline to illustrate the effectiveness of the framework under minimal assumptions. Extensions to more flexible temporal priors are explored later in Section 4.4.

4.2.1. Covariates Used in the Models

Table 2 provides a detailed overview of the covariates used across different modeling approaches. In micro-level models, policyholder-specific covariates (car age, class, fuel type, contract type, and region) are used to model the claim frequency intensity, while the reporting delay structure incorporates both policyholder covariates and time-dependent covariates. The time-dependent covariates differ between continuous and discrete models: In continuous-time models, we use the exact day of accident occurrence, while in discrete-time models, where data is aggregated monthly, we use the day on which the month ends. For both types of model, we incorporate the month on which the accident occurs as a covariate. For the IPW method, the same covariates are used for modeling reporting delay, but claim severity is additionally included, as the method applies only to reported claims. In the Bayesian model, policyholder characteristics are replaced with portfolio-level summary statistics—averages for continuous variables (e.g., car age) and proportions for categorical variables (e.g., car class distribution). Time-dependent covariates remain the same as in the discrete-time micro-level models, ensuring consistency across approaches.

TABLE 2
Covariates Used in Different Modeling Approaches

Covariates	Micro-level models	IPW method	Bayesian model
Policyholder-specific covariates			
Age of driver	√	√	Average age in portfolio
Age of car	√	√	Average age in portfolio
Car class (A, B, C)	√	√	Percentage of each class
Fuel type (gasoline/diesel)	√	√	Percentage of each type
Contract type (new/renewal)	√	√	Percentage of renewals
Region (five regions)	√	√	Percentage of each region
Time-dependent covariates			
Month of accident occurrence	√	√	√
Day of accident occurrence	√ (continuous-time only)	√	–
End-of-month day (for monthly aggregation)	√ (discrete-time models)	–	√
Additional covariates			
Claim severity	–	√ (since applied to reported claims)	–

4.2.2. Adjustment for Maximum Reporting Delay and Performance Comparison

A key consideration in evaluating the discrete-time micro-level models and the Bayesian models is that they impose a maximum reporting delay of $D = 9$ months. Empirically, the longest observed reporting delay in the data is 24 months, which means that models truncated at $D = 9$ inherently underestimate the total number of claims. However, since more than 99% of claims are reported within 9 months, the magnitude of this underestimation is negligible.

To ensure a fair comparison with models that estimate the full IBNR, we supplement the discrete-time and Bayesian estimates with Chain Ladder estimates for the tail cells beyond 9 months, that is,

$$\{N_{t,d} : t + d > T,\ d > 9\}.$$

Specifically, this is done by applying the Chain Ladder development factors for $d > 9$ to the cumulative claim count triangle implied by the discrete-time and Bayesian models, thereby extending their predictions to the tail. Thus, the reported IBNR from these models consists of the fitted component with $d \leq 9$ plus the Chain Ladder estimate of the residual tail with $d > 9$.

Because the contribution of these tail cells is minimal relative to the main component, this adjustment has no effect on predictive accuracy or on the ranking of models. We nevertheless apply it to ensure that all models are evaluated against a consistent IBNR definition.

4.3. Results

IBNR claims count estimations were performed at the end of each month from January 2014 to December 2015, resulting in 24 valuation points. For each valuation date, the training set consists of all claims reported up to that valuation date. The models are fitted using this reported claims data to estimate the parameters, which are then used to predict the IBNR claim count.

Since our dataset extends to December 2017, and all claims (except for two) are reported within two years of occurrence, we have reliable actual IBNR claim counts to compare against the different models' estimates.

To evaluate the performance of each model, we calculate the absolute percentage error (APE) of the IBNR claim count point estimates at each of the 24 valuation points. The APE is defined as

$$\text{APE} = \left| \frac{\text{Estimated IBNR} - \text{Actual IBNR}}{\text{Actual IBNR}} \right|. \tag{6}$$

The mean and standard deviation of the APEs across the valuation points for each model are summarized in Table 3.

4.3.1. Bayesian Models' Results Analysis & Comparison with the Chain Ladder

As previously discussed, we explore two versions of the Bayesian framework: a baseline model, where covariates are incorporated but do not directly influence the delay structure, and a full model, where covariates explicitly influence the delay distribution by allowing delay-specific regression coefficients ($\boldsymbol{\eta}_d, \boldsymbol{\delta}_d$).

The results in Table 3 show that the baseline model yields only marginal improvement over the classical Chain Ladder, with mean APE decreasing from 15.55% to 14.37%. In this dataset, treating accident effects as fixed effects already provides stable estimates; the first-order random walk does not materially improve prediction accuracy (further analysis of alternative temporal structures is given in Section 4.4). Moreover, the incorporation of covariates without allowing them to affect the delay structure does not materially improve predictive performance either. Notably, the baseline model without the interaction term performs slightly better (14.16%), suggesting that these interaction effects may be unnecessary and potentially add noise in this dataset.

By contrast, the full model achieves a substantial improvement, reducing mean APE from 15.55% (monthly Chain Ladder) to 7.97%. Dropping the interaction term yields the best performance overall, with a mean APE of 7.89% and the lowest standard deviation of 6.3%. The key driver of this improvement is the *month of occurrence* covariate, but only when it is allowed to influence the delay distribution through delay-specific regression coefficients. In the baseline specification, where covariates enter only through a common effect across delays, this seasonal signal is not captured and no predictive gain is observed. Once delay-specific coefficients are introduced, the strong seasonality in reporting patterns becomes evident (see Figure 4). A reduced model including only this seasonal covariate with delay-specific effects already reduces the mean APE to 10.1%. Adding portfolio-level summaries yields further, though smaller, improvements. The limited impact of policyholder summary statistics reflects the relative stability of portfolio composition over the observation window.

Given the pronounced seasonality in our data, we also examined the performance of the Chain Ladder under different levels of temporal aggregation (annual, semiannual, quarterly, and monthly). As noted by Tee and Käärik (2022), finer aggregation generally improves Chain Ladder performance by retaining more information about the reporting process. In our dataset, however, yearly aggregation coincidentally produces lower error (mean APE of 12.69%) compared to semiannual (23.92%), quarterly (17.08%) and monthly (15.55%), because annual averaging effectively smooths out the seasonal fluctuations. Interestingly, the yearly Chain Ladder even outperforms the Bayesian baseline model. Once seasonality is explicitly modeled, however, as in the full Bayesian framework, finer aggregation regains its advantage and predictive performance improves substantially beyond both the yearly Chain Ladder and the baseline Bayesian specification.

In summary, two key points emerge. First, predictive gains are not automatic from more complex structures: Temporal smoothing and interactions alone did not outperform the Chain Ladder baseline, and in fact, excluding interactions slightly improved performance. Second, improvements arise primarily when covariates capture genuine systematic variation—in this case, strong seasonal reporting effects. This suggests that in other datasets, the value of additional model components will depend on whether meaningful variation is present in covariates such as seasonality or shifting portfolio risk composition.

TABLE 3
Model Performance Comparison: Mean Error and Standard Deviation

Model	Mean error	Standard deviation
Chain Ladder, monthly	0.1555	0.1102
Poisson, continuous	0.1077	0.0827
HMM, continuous	0.1125	0.0639
IPW method	0.1464	0.0886
Poisson, discrete	0.1414	0.0855
HMM, discrete	0.0944	0.0836
Bayesian baseline model	0.1437	0.0774
Bayesian baseline model, no interaction	0.1416	0.0837
Bayesian full model	0.0797	0.0674
Bayesian full model, no interaction	0.0789	0.0630

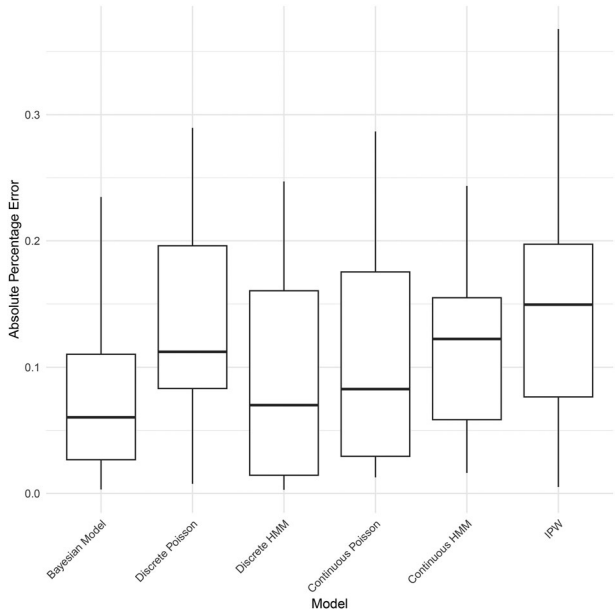


FIGURE 2. Distribution of Absolute Percentage Error (APE) across Models.

4.3.2. Comparison with Micro-Level Models

One of the most striking findings is that our Bayesian macro-level model outperforms all micro-level approaches. The full Bayesian model without the interaction term achieves the lowest mean APE (7.89%) and standard deviation (0.063), surpassing even the best-performing micro-level specifications.

This is noteworthy because micro-level models are often assumed to be superior in principle, given their full granularity. Our results show, however, that a carefully structured macro-level framework can deliver better predictive accuracy with fewer modeling complexities.

Figure 2 illustrates the distribution of APE across all models. The full Bayesian model has both the lowest median APE and the smallest interquartile range, highlighting its predictive stability. This suggests that macro-level Bayesian extensions can provide a robust and interpretable alternative for practitioners, while still maintaining the operational simplicity of aggregated models.

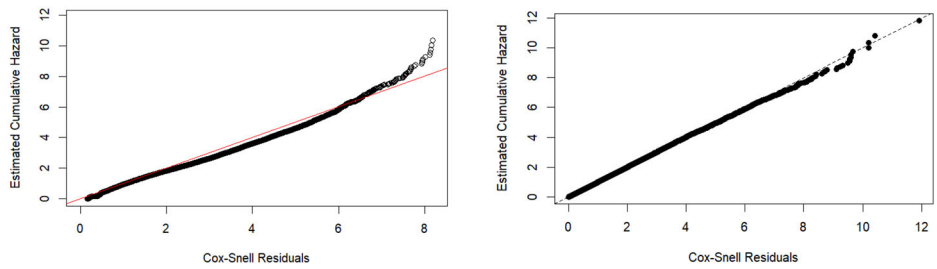
4.3.3. Sensitivity of Continuous-Time Models to Reporting Delay Specification

One critical drawback of continuous-time micro-level models is their sensitivity to the choice of distribution used to model reporting delays. While earlier results in this article adopted a log-logistic proportional hazards regression model—selected for its strong empirical fit—one might reasonably question whether a better fitting model for the reporting delay could improve the predictive accuracy of IBNR claim count estimates. To investigate this, we repeated the analysis for all continuous-time models requiring reporting delay specification, now using a Cox proportional hazards model instead.

Figure 3 presents the Cox–Snell residual plots for both the log-logistic and Cox regression models. These plots were generated using reporting delay models fitted to all claims that occurred before 2016. While both distributions exhibit a good fit, the Cox model shows a near-perfect alignment with the 45-degree line, suggesting an even better overall fit to the delay distribution.

Surprisingly, improvements in the fit of the reporting delay model did not consistently translate into better IBNR predictions. Table 4 compares the mean APE for three continuous-time models under the two specifications for the reporting delay distribution: Cox regression and log-logistic regression. The results for the IPW method and the continuous Poisson model are quite similar under both specifications, with the IPW method performing slightly better under the Cox regression and the Poisson model slightly worse. However, the continuous HMM model shows noticeably worse performance with the Cox regression, with its mean APE increasing from 11.3% under the log-logistic regression to 14.3%.

While the difference in mean error for the HMM model appears large, it is worth noting that the *median* mean APE across the 24 valuation dates is 12.3% for the log-logistic model and 13.0% for the Cox regression—indicating that the degradation is not as severe as the mean alone suggests.



(a) Log-logistic regression (b) Cox regression

FIGURE 3. Cox–Snell Residual Plots for the Log-Logistic and Cox Regression Models, Fitted to Reporting Delays of Claims Occurring before 2016.

TABLE 4
Mean Absolute Percentage Error under Different Reporting Delay Models

Model	Log-logistic	Cox regression
Continuous Poisson	0.108	0.118
Continuous HMM	0.113	0.143
IPW	0.146	0.133

To better understand this effect, we investigated the estimation accuracy of the integral

$$\int_{d_{T-1}}^{d_T} F_{U|\boldsymbol{x}_i,t}(\tau - t)dt,$$

which approximates the probability that a claim occurring in the final observation interval $[d_{T-1}, d_T)$ is reported before the valuation date τ. This quantity is crucial for IBNR estimation, as most unreported claims fall into this last interval. For each valuation date, we compared the model-estimated average of this integral across policyholders to the actual proportion of claims reported from that interval. The log-logistic regression produced a lower mean absolute deviation from the true proportions (1.7%) compared to the Cox regression (2.2%).

Importantly, the IBNR estimates tended to be more accurate on dates when the model-estimated average of this integral was closer to the true reporting percentage, whether this occurred under the log-logistic or the Cox regression model. In our dataset, the log-logistic model more often produced the closer estimates across the 24 valuation dates, which explains its overall better performance.

This sensitivity is particularly pronounced for the HMM-based model due to the "double effect" discussed in Section 2.2. In this model, estimating the number of IBNR claims involves two stages: (1) inferring the most likely sequence of hidden states using the Viterbi algorithm, and (2) simulating unreported claims conditional on those states. If the reporting probability integral is overestimated (underestimated), the model will tend to infer a lower intensity (higher intensity) hidden state for the final period, since it believes more (fewer) claims should have been observed. At the same time, the model also underestimates (overestimates) the complementary probability of a claim remaining unreported. These two sources of error compound, leading to a more severe underestimation (overestimation) of the IBNR count.

Overall, this subsection highlights the sensitivity of continuous-time micro-level models to the choice of reporting delay distribution—especially for models with time-dependent arrival processes. Even when one specification of the delay distribution fits the data more accurately, this does not guarantee better IBNR results. Accurate estimation of short-delay probabilities is particularly critical, and errors in these estimates can propagate nontrivially through the IBNR prediction pipeline.

4.3.4. Prediction Interval Analysis: Chain Ladder vs. Full Bayesian Model

A critical aspect of IBNR model evaluation lies not only in the accuracy of the point estimates but also in the credibility and reliability of the associated prediction intervals. Figure 4 presents the 95% prediction intervals for the IBNR claim count estimates obtained from the Bayesian implementation of the Chain Ladder model and the full Bayesian model. For each valuation date, the actual number of IBNR claims is also displayed, with red points indicating instances where the true value falls outside the interval and blue points indicating it lies within.

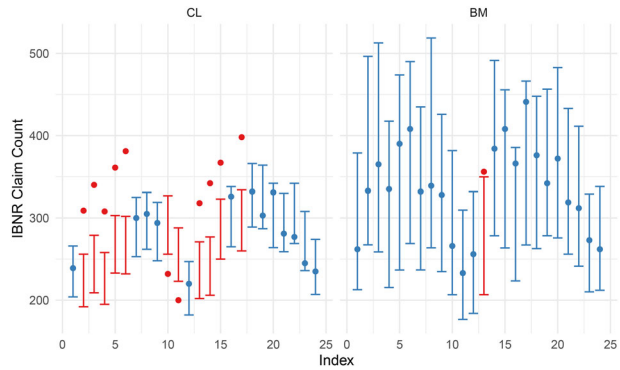


FIGURE 4. Error Bar Plots Showing 95% Prediction Intervals for IBNR Claim Count Estimates across Validation Dates. Left Panel: Chain Ladder (CL); Right Panel: Bayesian Model (BM). Blue Intervals Indicate That the Actual IBNR Claim Count Falls within the Interval, While Red Intervals Indicate Otherwise.

Both models are estimated in a Bayesian framework using `Stan`, allowing for direct sampling from the posterior predictive distribution of IBNR counts. The Chain Ladder model follows the Poisson formulation in Equation 2, where the expected claim count is decomposed as $\lambda_{t,d} = \alpha_t \beta_d$, with $\sum_{d=0}^{D} \beta_d = 1$. For identifiability, the delay factors are assigned a Dirichlet prior, and the development year effects α_t are given weakly informative log-normal priors to reflect multiplicative effects over time. Notably, the posterior means of the IBNR claim counts under this Bayesian formulation coincide with the estimates produced by the classical deterministic Chain Ladder method.

The plotted intervals reveal a clear contrast. The Bayesian Chain Ladder model produces relatively narrow intervals that frequently fail to contain the true number of IBNR claims across validation periods. In contrast, the full Bayesian model yields wider intervals that successfully capture the true values across all but one date. While the intervals from the full model may appear overly wide, their breadth reflects the genuine variability in the data. Importantly, they achieve near-complete coverage without compromising the accuracy of the point estimates—a level of performance unmatched by any other model in this study.

4.4. Alternative Temporal Structures

To illustrate the flexibility of the proposed framework, we also estimate a reduced model of the form

$$\log \lambda_{t,d} = \mu + \alpha_t + \beta_d + \gamma_{m(t),d},$$

where $\gamma_{m(t),d}$ captures seasonality by month of occurrence, and $m(t)$ denotes the month corresponding to accident period t. This reduced form is analogous to our full framework without the accident–delay interaction term, and retaining only the seasonal covariate. The parameters β_d and $\gamma_{m(t),d}$ are treated as fixed effects, while different temporal structures are imposed on the accident effects α_t.

We implement several temporal priors for α_t, including fixed effects, first- and second-order random walks, first-order autoregressive process, local linear trend, penalized splines, and Gaussian processes. For the Gaussian process, we adopt a squared exponential kernel, while for the penalized spline specification we use a basis with 12 knots and a second-order difference penalty. Moreover, we kept priors weakly informative across all specifications for comparability. Each of these alternatives can be incorporated by modifying only a few lines in the `Stan` script, and full implementations are available in our GitHub repository (https://github.com/HassanAbdelrahman/IBNR_GLM_Stan), allowing practitioners to adapt the models directly to their own datasets.

Table 5 reports predictive accuracy for each structure. For this dataset, fixed effects already perform reasonably well, with only modest improvements under certain dynamic specifications (e.g., the first-order random walk and autoregressive process). The Gaussian process prior performs best overall, achieving the lowest mean APE and standard deviation. We reran our full model without the interaction term using a Gaussian process prior on α_t; the mean APE is 7.66%, which is slightly better than the 7.89% obtained with the first-order random walk.

The main takeaway is that practitioners can readily experiment with different temporal assumptions within the same unified framework. Importantly, there is no need to worry about estimation details: `Stan` handles inference automatically once the model is specified, allowing practitioners to focus on model design rather than custom estimation routines.

TABLE 5
Predictive Performance under Alternative Temporal Structures for α_t

Specification	Mean APE	Std. of APE
Fixed Effects	10.72%	0.0805
First-order random walk	9.98%	0.0728
Second-order random walk	10.99%	0.0780
First-order autoregressive process	10.12%	0.0732
Local linear trend	10.80%	0.0764
Penalized splines	11.48%	0.0827
Gaussian process	9.72%	0.0727

TABLE 6
Average Runtimes (Minutes) for the Reduced Model under Different Temporal Structures

Temporal structure	Runtime (minutes)
Fixed effects	1.97
First-order random walk	1.44
Second-order random walk	13.01
First-order autoregressive process	4.92
Local linear trend	5.08
Penalized splines	3.67
Gaussian process	9.11

For datasets with fewer valuation points, predictive comparisons may be less reliable. In such cases, we recommend starting with a simpler temporal structure, such as the first-order random walk or AR(1), which generally perform well and are less prone to overfitting. More flexible priors (e.g., splines or Gaussian processes) can be explored as additional data become available.

4.5. Computational Considerations

Table 6 reports the average runtime (in minutes) of the reduced model under the different temporal priors for α_t considered in Section 4.4. All models were fitted on the University of Toronto Department of Statistical Sciences server (64-core AMD EPYC 7702 P, 2 GHz), with each script executed on a single core. The reported times correspond to fitting the model at the first valuation date (January 2014).

The runtimes are very reasonable; most reduced models complete in under 5 minutes, with only the second-order random walk requiring longer (13 minutes). Even the more computationally intensive Gaussian process model converges in under 10 minutes.

For completeness, the full model requires somewhat longer fitting times: 13.69 minutes without interaction terms and 24.32 minutes with interactions.

Overall, these runtimes underscore the practical feasibility of the proposed framework. Even complex specifications can be estimated within minutes on standard hardware, making the approach attractive for practitioners who wish to fit and compare alternative temporal structures.

5. DISCUSSION OF MODEL TRADE-OFFS

The models used in the analysis illustrate a sequence of modeling trade-offs that involve four key aspects: the level of data aggregation, the granularity of modeling (policyholder vs. portfolio), the temporal structure imposed on the claim arrival process, and whether claim occurrence and reporting delay are modeled jointly. Each modeling decision enhances some features at the expense of others, and the relative performance of each approach depends on which of these aspects is prioritized.

We begin with the IPW method. This approach models only the reporting delay, without requiring an explicit model for the claim occurrence process, since IBNR claim counts can be estimated by adjusting the observed claims using fitted delay probabilities. While the method can theoretically yield unbiased estimates, it treats the delay distribution in isolation. In practice, however, the number of claims reported in a given period reflects both the frequency of underlying claim occurrences and the speed of reporting. As such, meaningful inference often requires modeling the arrival and delay processes jointly. In our experience, predictions from the IPW method can also become unstable if the delay distribution is even slightly misspecified.

Micro-level continuous-time models aim to capture both claim occurrence and reporting delay by modeling events at the level of individual policyholders using the full time resolution of the data. Typically, the claim arrival process is modeled as a Poisson process with an intensity that depends on individual-level covariates. A key advantage of this approach is its ability to differentiate risk across individuals using granular information. However, modeling a flexible temporal structure for claim frequency at this level is extremely challenging. Most studies in the literature rely solely on time-varying covariates to account for trends over time. A notable exception is the model by Abdelrahman et al. (2025), which introduces a latent HMM to govern the claim intensity process across all policyholders, thereby embedding an implicit portfolio-level temporal structure. Still, these models suffer from a structural limitation: The reporting delay distribution is estimated independently of the unobserved claim arrivals. This disconnect introduces estimation challenges and limits the model's ability to capture the joint dynamics between frequency and delay.

Discrete-time micro-level models offer a natural solution to the problem of coherence. By aggregating time into intervals (e.g., weeks or months), these models allow for the joint modeling of claim occurrence and reporting delay within a single likelihood. However, the challenge of modeling a flexible temporal structure persists. Because claim frequency is still modeled at the policyholder level, incorporating rich temporal dynamics remains difficult.

These limitations motivate a different direction—toward models that operate at the portfolio level. The Bayesian model used in the analysis exemplifies this approach. It does not model policyholders individually but instead incorporates their risk characteristics through portfolio averages and scales claim intensity using exposure. In doing so, it sacrifices individual-level granularity but gains the ability to specify a much more flexible temporal structure for the claim arrival process. Importantly, the model jointly captures claim frequency and reporting delay, and produces fully coherent posterior inference.

It is worth emphasizing that this model does not ignore policyholder risk—rather, it takes a middle ground between policyholder- and portfolio-level modeling. It assumes that all insureds share the same risk profile given by the empirical averages of the individual characteristics, a simplification that proves effective for IBNR count prediction.

To understand why, consider how IBNR claims are estimated in micro-level models. Given a fitted likelihood, IBNR claim counts are typically obtained through simulation, where claim occurrence for each policyholder i is generated from a Poisson process with intensity

$$\lambda(t;\boldsymbol{x}_i)\big(1 - F_{U|t,\boldsymbol{x}_i}(\tau - t;\boldsymbol{x}_i)\big),$$

where $\boldsymbol{x}_i$ represents the policyholder's risk characteristics. However, in most insurance applications, the claim occurrence rate $\lambda(t;\boldsymbol{x}_i)$ is close to zero across policyholders. Even if the reporting delay distribution $F_{U|t,\boldsymbol{x}_i}$ varies across individuals, its impact remains minor when multiplied by a near-zero claim intensity.

As a result, simulating IBNR claim counts at the policyholder level effectively reduces to a random sampling mechanism, where individual claims are drawn in a manner that reflects overall portfolio risk. Since the final IBNR count is obtained by averaging over multiple simulations, the estimate depends primarily on the portfolio-level characteristics rather than individual-level variations. This suggests that for IBNR count prediction, incorporating policyholder averages is sufficient, as the estimation ultimately reflects the portfolio's overall risk profile.

In conclusion, the analysis highlights a central trade-off in IBNR modeling. Moving from policyholder-level to portfolio-level modeling entails a loss in granularity but enables richer temporal structures and coherent joint modeling. Each modeling approach has its strengths and limitations, and these trade-offs suggest several avenues for future research—such as developing hybrid models that retain individual-level heterogeneity while allowing for flexible temporal dynamics, or devising practical estimation strategies for coherent micro-level models that jointly capture claim occurrence and delay.

6. CONCLUSION

This article demonstrates that substantial improvements in IBNR claim count estimation can be achieved without opting for fully granular micro-level models. Our proposed Bayesian extension of the classical Chain Ladder retains its aggregate structure while introducing a highly flexible and general modeling framework. It accommodates portfolio-level risk characteristics,

seasonal patterns, and interaction terms, and allows practitioners to impose different temporal structures—such as random walks, autoregressive processes, splines, or Gaussian processes—on the accident and delay effects. Using real-world auto insurance data from a European insurer, we benchmarked our approach against several micro-level models from the literature. The results showed that our model not only matched but outperformed these more complex alternatives.

A key strength of the proposed approach is its practicality. Thanks to modern probabilistic programming tools like `Stan`, practitioners can fit these models without having to deal with the complexities of custom estimation routines. To further support adoption, we provide fully reproducible code for all specifications—including different temporal priors—so that practitioners can readily adapt the models to their own datasets.

By combining the interpretability and accessibility of the Chain Ladder with the flexibility and rigor of modern Bayesian methods, this framework bridges the gap between academic innovation and industry practice. It offers insurers a reliable, interpretable, and computationally manageable tool for IBNR claim count estimation that can easily evolve with their data and needs.

A natural direction for future work is to extend our Bayesian temporal modeling framework beyond claim counts to also cover claim severities. By modeling both components—claim frequency and claim severity—within a unified Bayesian framework, we aim to provide separate and coherent estimates of the RBNS and IBNR claim reserves in line with the Collective Reserving Model proposed by Wahl et al. (2019). This would yield a comprehensive approach to reserving that remains grounded in aggregated data, yet leverages flexible temporal structures and incorporates portfolio-level risk characteristics. Furthermore, by adopting a Bayesian perspective, we can relax some of the restrictive assumptions in the Collective Reserving Models and introduce more realistic dependence structures, which we believe can lead to improved predictive performance and more robust reserve estimates.

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APPENDIX A. IMPLEMENTATION GUIDE FOR THE PROPOSED MODEL

This appendix provides a practical guide to fitting the proposed model described in Section 3 using `cmdstanr`, an R interface to `Stan`. We break down the workflow into three parts: (1) data preparation and model fitting, (2) `Stan` model specification, and (3) computation of the posterior distribution of IBNR claims count.

A.1. DATA PREPARATION AND MODEL FITTING

We begin by preparing the input data from the run-off triangle. The triangle is assumed to be provided as a data frame or matrix with T rows (accident periods $t = 1, ..., T$) and $D + 1$ columns (reporting delays $d = 0, ..., D$), where the cell (t, d) contains the observed number of claims $N_{t,d}$. The actual evaluation time T is denoted `n_periods` (to avoid conflicts with the Boolean `TRUE`/`FALSE`), and the maximum delay D is stored as `max_delay`.

Claims with $t + d > T$ are censored to reflect that they are not observable at evaluation time T:

```
censored_triangle < - run_off_triangle
censored_triangle[outer(1:n_periods, 0:max_delay, FUN = "+") > n_periods] < - NA
```

Next, we convert the triangle into long format suitable for `Stan` using a custom reshaping function:

```
reshape_triangle < - function(triangle_matrix) {
  data.frame(
    N = as.vector(as.matrix(triangle_matrix)),
    Period = rep(1:nrow(triangle_matrix), times = ncol(triangle_matrix)),
    Delay = rep(0:(ncol(triangle_matrix) - 1), each = nrow(triangle_matrix))
  )
}
```

This yields a data frame where each row corresponds to a unique (t, d) combination, with three columns: the claim count $N_{t,d}$ (column `N`), the accident period `Period`, and the delay `Delay`.

We then append exposure, where the exposure for each accident period t is repeated $D + 1$ times:

```
data < - reshape_triangle(censored_triangle)
data$exposure < - rep(exposure[1:n_periods], max_delay + 1)
```

In the code just shown, `exposure` is a vector reflecting the exposure w_t for $t = 1, ..., T$. We then split the data into observed and unobserved parts, keeping only observations with $t + d \leq T$:

```
observed_data < - data % > % filter(!is.na(N))
unobserved_data < - data % > % filter(is.na(N))
```

Finally, we construct the list of inputs for `Stan` and run the model:

```
stan_data < - list(
  N_obs = nrow(observed_data),
  L = n_periods,
  D = max_delay + 1,
  n = observed_data$N,
  exposure = observed_data$exposure,
  period = observed_data$Period,
  delay = observed_data$Delay,
  covariates = covariates,
  p = ncol(covariates)
)

mod < - cmdstan_model("GLM_Model.stan")
fit < - mod$sample (
  data = stan_data,
  seed = 123456,
  refresh = 0,
  show_messages = FALSE,
  show_exceptions = FALSE
)
```

Here, the covariates are the time-specific and policyholder-level summary statistics (i.e., $\boldsymbol{X}_t$ and $\boldsymbol{Z}_t$), combined into a covariate matrix with `n_periods` rows representing the values observed at each accident period.

A.2. `STAN` MODEL SPECIFICATION

Shown next is the complete `Stan` model used to fit the Bayesian GLM described in Section 3. The model takes as input the observed claim counts $N_{t,d}$ (denoted `n`), the associated accident period and delay indices (period and delay), the

exposure w_t, and a matrix of covariates summarizing policyholder and temporal features. The core structure of the model reflects a log link function with random effects and time-varying covariate coefficients.

We use weakly informative priors for the parameters in our analysis; however, users may specify any appropriate prior distributions based on their context or data. The following `Stan` code specifies the priors we used in the empirical results reported in Section 4:

```
// The input data
data {
  int<lower = 0>N_obs;
  int<lower = 1>L;              // number of accident periods
  int<lower = 1>D;              // number of reporting delays
  int<lower = 1>p;              // number of covariates

  array[N_obs] int<lower = 0>n;                  // claim counts
  array[N_obs] real<lower = 0>exposure;            // exposure
  array[N_obs] int<lower = 0, upper=L> period;      // accident period (0-based)
  array[N_obs] int<lower = 0, upper=D> delay;      // reporting delay (0-based)
  matrix[L, p] covariates;                        // covariates at period level
}

// The parameters
parameters {
  real mu;
  vector[L] alpha_raw;
  vector[D] beta_raw;
  matrix[L, D] gamma_raw;
  matrix[p, D] delta;

  real<lower = 0>sigma_alpha;
  real<lower = 0>sigma_gamma;
  real<lower = 0>phi;
}

// Transformed parameters
transformed parameters {
  vector[L] alpha;
  vector[D] beta;
  matrix[L, D] gamma;
  vector[N_obs] log_lambda;

// Enforce sum-to-zero constraints
alpha=alpha_raw - mean(alpha_raw);
beta=beta_raw - mean(beta_raw);
for (d in 1:D)
  gamma[, d] = gamma_raw[, d] - mean(gamma_raw[, d]);

// Log-rate for each observation
for (i in 1:N_obs) {
  log_lambda[i] = mu
            + alpha[period[i]]
            + beta[delay[i]+1]
            + gamma[period[i], delay[i]+1]
            + covariates[period[i], ] * delta[, delay[i]+1]
            + log(exposure[i]);
  }

}
```

```
// The model
model {
  // Likelihood
  n ~ neg_binomial_2_log(log_lambda, phi);

  // Priors
  phi ~ exponential(0.1);
  mu ~ normal(0, 1);

  // Smooth prior on alpha_raw (random walk + sum-to-zero)
  alpha_raw[1] ~ normal(0, sigma_alpha);
  for (t in 2:L)
    alpha_raw[t] ~ normal(alpha_raw[t - 1], sigma_alpha);
    sigma_alpha ~ normal(0, 0.1);

    // Independent prior on beta_raw (sum-to-zero will apply)
    beta_raw ~ normal(0, 1);

    // Smooth prior on gamma_raw by row (over accident period, for each delay)
    for (d in 1:D) {
      gamma_raw[1, d] ~ normal(0, sigma_gamma);
      for (t in 2:L)
        gamma_raw[t, d] ~ normal(gamma_raw[t - 1, d], sigma_gamma);
}
sigma_gamma ~ normal(0, 0.1);

// Independent prior on delta
for (k in 1:p)
for (d in 1:D)
  delta[k, d] ~ normal(0, 1);
}
```

This model is implemented in GLM_Model.stan and can be compiled and fitted using the cmdstanr code provided in the previous section. In the next section, we show how to extract the posterior distribution of the IBNR claims count from the fitted model.

A.3. POSTERIOR DISTRIBUTION OF IBNR CLAIMS COUNT

Once the model is fitted, we can extract posterior draws and compute the posterior distribution of the total IBNR claims count. The following R code demonstrates this process using the posterior samples stored in fit, and the unobserved part of the triangle prepared earlier (named unobserved_data).

```
draws_df < - fit$draws(format = "df") % > % as_tibble()
posterior_ibnr_samples < - rep(0, 4000) # assumes 4000 posterior draws
for (i in 1:nrow(unobserved_data)) {
    period < - unobserved_data$Period[i]
    delay < - unobserved_data$Delay[i]
    exp_i < - unobserved_data$exp[i]

    t < - paste0("alpha[", period, "]")
    k < - paste0("beta[", delay + 1, "]")
    s < - paste0("gamma[", period, ",", delay + 1, "]")
    p < - paste0("delta[", 1:ncol(covariates), ",", delay + 1, "]")

    log_lambda < - draws_df$mu +
                   draws_df[[t]] +
                   draws_df[[k]] +
```

```
                    draws_df[[s]] +
                    as.matrix(draws_df[, p]) %*%
                    t(as.matrix(covariates[period, drop = FALSE])) +
                    log(exp_i)

log_lambda < - as.numeric(log_lambda)
theta < - draws_df$phi

posterior_ibnr_samples < - posterior_ibnr_samples +
  rnegbin(n = length(log_lambda),
        mu = exp(log_lambda),
        theta = theta)
}
```

This loop computes the posterior distribution of the total IBNR claims count by simulating unobserved claims for each censored cell (t, d) using the negative binomial distribution implied by the posterior draws. The resulting vector `posterior_ibnr_samples` contains posterior samples of the total IBNR claims count.

A.4. DATA PREPARATION CHECKLIST

The code provided in the precedingassumes that the user has prepared the following three objects with these exact names:

- `run_off_triangle`: A $T \times (D+1)$ matrix or data frame, where rows correspond to accident periods ($t = 1, \ldots, T$) and columns correspond to reporting delays ($d = 0, \ldots, D$). Each cell (t, d) contains the observed claim counts $N_{t,d}$.
- `covariates`: A $T \times K$ data frame, where each row corresponds to an accident period and each column represents a covariate summarizing portfolio characteristics (e.g., drivers' average age) or a time-dependent covariate (e.g., month of occurrence).
- `exposure`: A numeric vector of length T representing the exposure w_t for each accident period t.

Practitioners simply need to ensure that these objects are created with the specified names and that their dimensions are consistent with the data before executing the scripts.

To guide users in creating these objects from their own portfolio and claims data, we provide a small simulation study that demonstrates the complete process of generating a run-off triangle, exposure vector, and covariate matrix. The R script for this simulation study is available at https://github.com/HassanAbdelrahman/IBNR_GLM_Stan.